

## 1 DEVELOPMENT → TRAIN

**Disjoint pool**  
no final labels

**Validate contract**

modes · templates · parser  
retries · throughput · cost

**Cellwise cap**

1024 if all  $\geq 99\%$   
else 2048 if all  $\geq 99\%$   
else STOP

**Freeze D/R packages**

manifest before every label

**TRAIN**

K draws/package  
fit methods

## 2 TUNING

**Disjoint tune set**

fixed 1/6 family weights

**Calibrate + search**

same gate family, budget  
resources, tie-to-direct

**Select one**

candidate per  
learned method

**Freeze candidate hash**

no audit-set candidate change

## 3 HELD-OUT DEPLOYMENT AUDIT

**Frozen candidate**

one audit; no retuning

**Empirical-Bernstein**

score LCB · cost UCB  
spoilage Z UCB  
denominator B LCB

**All four  
checks pass?**

Level-gamma intersection–union pass rule for one frozen candidate  
Empirical safeguard—not conformal, pointwise, or shift guarantee

## 4 DEPLOYMENT / TEST

**New input x**

input only

**Frozen gate**

enabled only on pass  
otherwise bypassed

**Exactly one answer call**

D: direct or R: bounded  
native thinking

**TEST: offline policy swap only**

after action log; never changes deployment gate

PASS: enable frozen gate

FAIL: force D and bypass gate